

Low-Temperature SLID-TSV Interconnects for 3-D (MEMS) Packaging

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Abstract—Three-dimensional heterogeneous integration is becoming increasingly important in advanced packaging as device functionalities expand within smaller spaces. Three-dimensional interconnects such as through silicon via (TSV)-solid-liquid interdiffusion (SLID) interconnects offer a promising approach for achieving miniaturization, high integration, and reduced power consumption. However, well-known Cu–Sn SLID-TSVs require high bonding temperatures, leading to residual stress and cracks. This research focuses on developing 3-D interconnects by using Cu–Sn–In/Co SLID-TSVs, which decrease bonding temperatures and reduce these issues. Finite element (FE) simulations qualitatively compared stress states in both SLID-TSV systems, showing lower residual stress in the Cu–Sn–In/Co SLID system than in Cu–Sn SLID. The Cu–Sn–In/Co SLID-TSV underwent microstructural analysis and reliability tests, including high-temperature storage (HTS), thermal shock (TS), and tensile strength testing. Most samples were free of voids and cracks, with a few showing minor defects along the bond line after TS. No cracks were observed inside the Si and TSVs. This indicates that adopting the Cu–Sn–In/Co system and reducing the bonding temperature to 200 °C can effectively prevent crack formations across bond lines, Si, and TSVs. Furthermore, all the samples meet the tensile strength requirements according to MIL-STD-883 method 2027.2, with the highest value observed for HTS-tested samples. Hence, low-temperature 3-D SLID-TSV interconnects were successfully demonstrated, showing strong potential for 3-D MEMS-ICs.

Index Terms—3-D interconnects, heterogenous integration, low-temperature bonding, MEMS integration, solid-liquid interdiffusion (SLID)-through silicon vias (TSVs), tensile test, thermomechanical reliability.

I. INTRODUCTION

THREE-DIMENSIONAL heterogeneous integration is a transformative approach in the semiconductor industry, pushing the boundaries of device performance beyond the limitations of Moore’s law [1]. In this context, utilizing 3-D interconnects, provides a route to achieve miniaturization,

high integration, decreased power consumption, and improved reliability [2]. This involves establishing electrical connections within the substrate using through silicon vias (TSVs), which are then linked to electrically conductive interconnects. The physical stacking of devices can be achieved in diverse configurations such as wafer-to-wafer, die-to-die, and die-to-wafer bonding [3]. When considering the 3-D integration of MEMS and ICs, wherein separate chips and/or wafers are vertically combined, vertical interconnects or bonded metal bumps are employed for electrical connection. In addition, sealing rings are commonly employed for establishing the required vacuum level or controlled environment [4], [5].

Considering the evolving needs in advanced 3-D integration interconnects must meet various criteria, including the rising demand for low-temperature, narrow-pitch, and short-time bonding methods [6]. In addition, interconnects need to withstand subsequent 3-D integration steps without remelting or degradation [7]. Although bonding techniques such as adhesive bonding and solders can achieve bonding temperatures below 200 °C [8], [9], [10], [11], they do not meet other critical requirements, such as narrow pitch sizes, phase stability, or hermeticity. Given these requirements, solid-liquid interdiffusion (SLID) bonding and hybrid bonding emerge as promising techniques for advanced 3-D integration [12], [13], particularly in 3-D MEMS-ICs integration. However, hybrid bonding involves complex, costly, and time-consuming treatments, such as surface preparation, and may be suitable only for extremely small pitch sizes (below 4 μm). For most applications, SLID bonding, with a pitch size of at least 10 μm , is generally applicable. SLID bonding techniques provide joints with a stable microstructure and the capability for miniaturization down to a 10- μm pitch size [14] or even smaller. Furthermore, utilizing SLID bonding allows for the use of the same bonding materials and process for both interconnects and sealing rings [4], ensuring consistent performance under thermal, mechanical, and thermomechanical stresses. This capability can be achieved through the simultaneous creation of sealing rings and vertical interconnects via wafer-level bonding.

Cu–Sn SLID has been extensively researched and is recognized for its applications in small microbump production and MEMS integration [12], [15], [16], [17], [18]. Two intermetallic compounds (IMCs), Cu_3Sn and Cu_6Sn_5 , are observed in Cu–Sn SLID depending on the Cu and Sn thickness ratio, bonding temperature, and bonding time [19]. To ensure a thermodynamically stable stack with Cu remaining after bonding, the preferred structure is Cu/ Cu_3Sn /Cu [20]. This

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configuration is attainable at higher bonding temperatures compared to the bond line formation with Cu_6Sn_5 [21].

However, lowering the bonding temperature or shortening the bonding time may lead to Cu_6Sn_5 formation. Over time, this phase can undergo a polymorphic phase transformation from hexagonal to monoclinic and eventually transform into the Cu_3Sn phase. Both phase transformations introduce stresses to the system [22], [23], [24]. Conversely, studies by Tiwary et al. [25] suggest that higher bonding temperatures in the SLID system result in increased stress levels in the IMCs during cooling down. They successfully demonstrated an average >30% stress level reduction by lowering the bonding temperature from 320 °C to 200 °C. Therefore, it is preferred to utilize a TSV-SLID with a lower bonding temperature and a stable microstructure after bonding for 3-D MEMS-IC integration. This can be achieved by substituting Sn with low melting point solders such as Sn–In [26] and [27]. In addition, Cu–Sn SLID materials are often deposited using wet-chemical electroplating processes, which can complicate integration on wafers containing MEMS devices [28]. To address these challenges, one approach involves physically depositing a contact metallization on MEMS device wafers, such as using Co as suggested by Emadi et al. [28] for Cu–Sn-based SLID interconnects. This contact metallization interfaces with Cu–Sn SLID-deposited cap wafers and simplifies the process by eliminating the need for wet-chemical processes on the MEMS wafers.

This study aims to utilize TSV wafers bonded to MEMS wafers using the newly introduced Cu–Sn–In/Co SLID system by Emadi et al. [26]. This SLID system offers advantages such as avoiding electroplating on MEMS device wafers, lowering bonding temperature to 200 °C with $(\text{Cu}/(\text{Cu},\text{Co})_6(\text{Sn},\text{In})_5/\text{Co})$ with stable microstructure, and potentially reducing residual compression stress. The objective is to demonstrate the feasibility of employing TSV-low-temperature SLID interconnects in MEMS integration, ensuring defect-free interconnects without crack propagation through the silicon or around the TSVs. Furthermore, the study seeks to assess the thermomechanical reliability of TSV-SLID through high-temperature storage (HTS), thermal shock (TS), and tensile strength measurements.

II. PREPARATION MATERIALS AND METHODS

A. Specimen Preparation

Bonded wafers were divided into two distinct types: dummy device wafers, intended to house MEMS devices, and cap wafers [see Fig. 1(a)]. The cap wafers were fabricated on thermally oxidized (300-nm SiO_2), one-sided polished 200-mm Si(100) wafers manufactured by Okmetic. These wafers featured poly-Si-filled TSVs and various types of cavities designed for devices such as gyroscopes. The TSVs were rectangular in shape, measuring 15 μm by 50 μm from the top view of the wafers, with depths ranging from 150 to 170 μm . The wafers were reworked by Disco, from 200 to 150 mm. The thickness of the SLID metals and the bonding conditions were determined based on our previous studies [26], [29], [30], achieving a successful Cu–Sn–In/Co SLID bond with a single phase $(\text{Cu},\text{Co})_6(\text{Sn},\text{In})_5$ in the bond line. A metallization stack

consisting of a 400-nm Al contact layer, a 60-nm-thick TiW adhesion layer, and a 100-nm Cu seed layer was sputtered onto the wafer. Subsequently, a thick photoresist mask featuring ring and microbumps structures (with TSVs and microbumps centered) was developed through the lithography process, utilizing AZ15nXT Photoresist. A 5- μm copper layer was electroplated into the resist openings using the NB Semi plate Cu 100 bath, followed by sequential electroplating of 1.7- μm tin using the NB Semi plate Sn 100 solution and 1.7- μm indium using an indium sulfamate plating bath. Device wafers were prepared on thermally oxidized (300-nm SiO_2), double-sided polished 150-mm Si (100) wafers with a thickness variation (TTV) below 2 μm . A 400-nm-thick Al contact metallization layer was sputtered on the Si wafers. Then, a photoresist mask with ring and microbump structures was formed via lithography using AZ 5214 E Image Reversal Photoresist. A 60-nm-thick Ti adhesion layer and a 400-nm-thick Co layer were sputtered onto the Si wafer. Then, a protective thin layer of Au (10 nm) was sputtered onto the Co layer to prevent oxidation. Prior to bonding, the metallization stack on the device wafers was patterned using a liftoff process, while the cap wafers were patterned by removing the photoresist and etching away the Cu seed layer and TiW adhesion layer. Then, the bonding process was carried out using a bond aligner EVG 610 and a wafer bonder EVG 510. The wafers were aligned and then preheated to 100 °C. Then, the wafers were brought into contact using a 5-kN uniaxial contact force from the bottom plate and then heated to 200 °C at a heating rate of 10 °C/min. After holding the temperature at 200 °C for 1 h, the contact force was gradually released to zero and the temperature was gradually reduced to 65 °C at a cooling rate of 1 °C/min before the bonded pair was removed from the chamber. Subsequently, the bonded wafers were diced into 10 × 10 mm chips containing one sealing ring and few TSV-SLID interconnects. These chips were used in TS, HTS, and tensile tests, as well as in cross-sectional analysis. Fig. 1 gives a schematic illustration detailing the wafer's preparation and features, wafer alignments with a cross-sectional view depicting a section of aligned wafers, and the temperature and pressure profiles during bonding.

B. Thermal Treatment and Tensile Test

The HTS test was carried out for a minimum of 15 chips. The testing was conducted using a Heraeus Instruments Oven for a duration of 1000 h at a temperature of 150 °C. The TS test was conducted using the ESPEC TSA-71 S TS chamber system. The TS test was performed according to the JEDEC JESD22-A104D standard, with test condition G and soak mode 3. The TS test parameters were as follows. An operational temperature range of –40 °C to +125 °C, a ramp rate of 33 °C/min, a 10-min dwell time applied to both high and low temperatures, and a total cycle time of 30 min. A minimum of 15 samples were subjected to 1000 cycles. Then, the tensile strength of the interconnects for all as-bonded (AB), thermal-shocked, and HTS-tested samples was assessed using a stud pull approach. An MTS 858 Table System equipped with a Flex Test 40 Digital controller and an MTS Silent Flow HPU system was utilized for this purpose. Further details on the

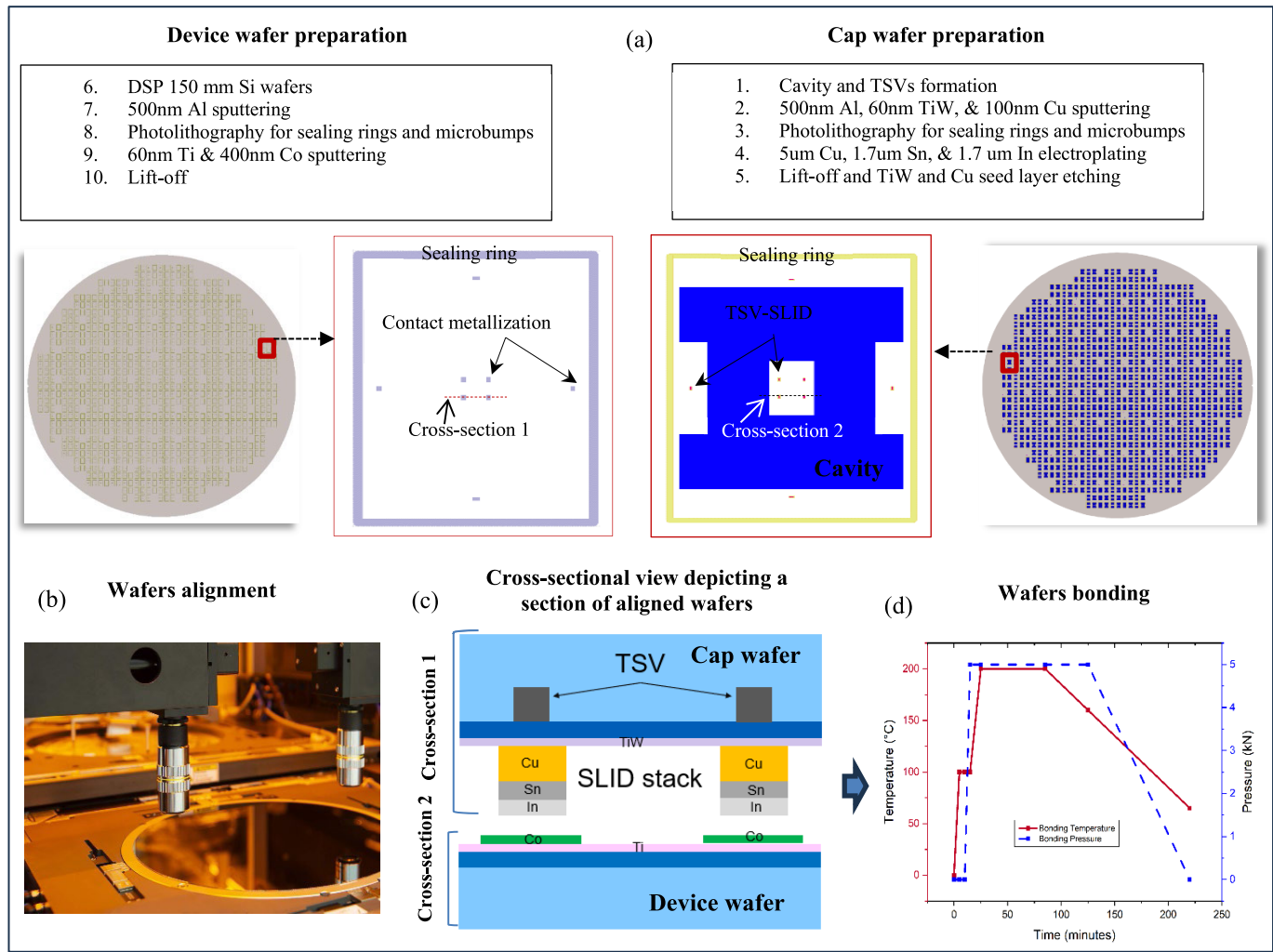


Fig. 1. Schematic of (a) wafers preparation, (b) wafers alignments, (c) cross-sectional view depicting a section of aligned wafers, and (d) bonding profile.

test setup can be found in [30]. A strain rate of 0.1 mm/s was applied during testing. A minimum of ten samples from each group of AB, TS-tested, and HTS-tested specimens were evaluated.

C. SEM/EDX Analysis

The samples were prepared for cross-sectional analysis using standard metallographic methods. The cross-sectional and fracture surfaces were analyzed using a Zeiss Sigma VP scanning electron microscope (SEM) equipped with an Ultim Max energy dispersive spectroscopy (EDS) detector. The composition of the phase was determined by averaging measurements from a minimum of five locations using EDX.

D. FE Simulation

Finite element (FE) simulations were performed to qualitatively study the thermomechanical stress distribution during the cooling down step of the bonding process. The FE simulations were carried out in COMSOL utilizing a solid mechanics physics interface. The plasticity domain was assigned to the Cu

and Al layers. All other domains were selected as linear elastic material. Fig. 2 shows a schematic of the model. The target of FE simulations is to study the thermomechanical stresses in the individual TSV-SLID bond. The stresses formed in the silicon region are not expected to extend beyond tens of micrometers from the bonds [31]. Therefore, in the model, the thickness of the Si layer is assumed to be 200 μm , exceeding the TSV structure by 25 μm , as illustrated in Fig. 2(a). In addition, since the thickness of adhesion layers is in the range of tens of nanometers, the TiW layer is not included in the model. The thickness and dimensions of all other layers in the model were consistent with the bonded samples used in the experiments. Fine mesh elements were assigned to the IMC, Cu, Co, SiO_2 , and Al domains, while normal mesh elements were assigned to the other domains.

The material physical parameters for poly-Si, SiO_2 , and Co domains were set to the default values available in the COMSOL library. The parameters for Cu and Si domains were gathered from the literature [25]. The initial yield stress of 275 MPa and the isotropic tangent modulus of 0.7 GPa were used for Al to account for its plastic material properties [32]. Young's modulus and Poisson's ratio for Cu_3Sn

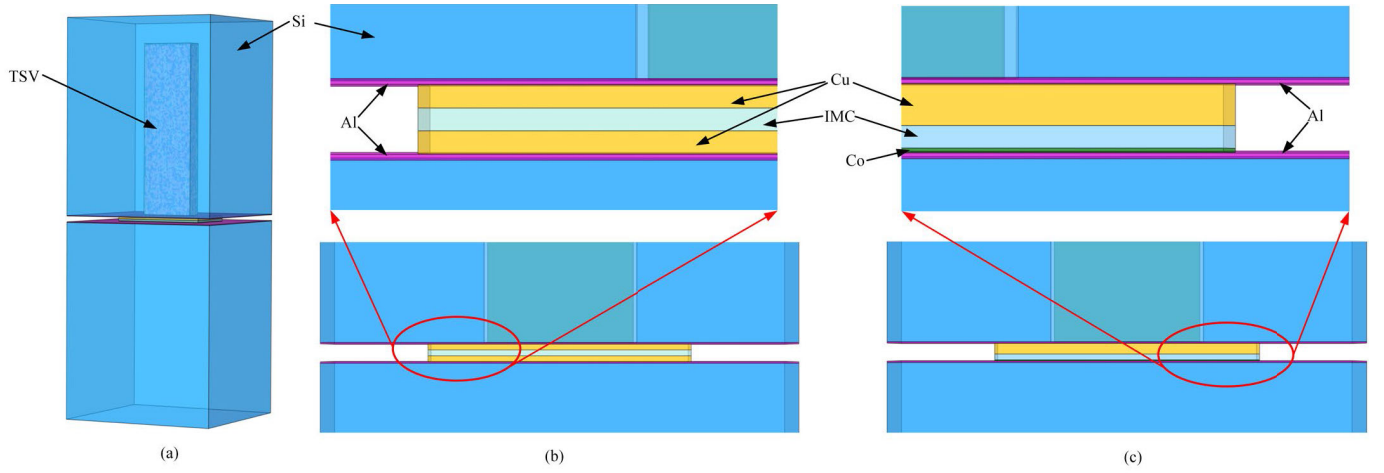


Fig. 2. Schematic of FE model (a) in 3-D with Si ($150 \times 160 \times 200 \mu\text{m}$) and TSV ($17.4 \times 52.4 \times 175.2 \mu\text{m}$); (b) Cu_3Sn as IMC, in cross section, Al ($150 \times 160 \times 0.5 \mu\text{m}$), Cu ($57 \times 90 \times 2 \mu\text{m}$), and Cu_3Sn ($57 \times 90 \times 2 \mu\text{m}$); and (c) Cu_6Sn_5 as IMC, in cross section, with Al ($150 \times 160 \times 0.5 \mu\text{m}$), Cu ($57 \times 90 \times 3.65 \mu\text{m}$), Cu_6Sn_5 ($57 \times 90 \times 2 \mu\text{m}$), and Co ($150 \times 160 \times 0.35 \mu\text{m}$).

TABLE I
MATERIALS PHYSICAL PROPERTIES

Materials	Domain	Density(kg/m^3)	Coefficient of thermal expansion ($1/\text{K}$)	Poisson's ratio	Young's modulus (GPa)	Initial yield stress (MPa)	Isotropic tangent modulus (GPa)
Cu	Plastic	8960	16.5×10^{-6}	0.34	120	250	2
Al	Plastic	Fig. 7(a)	Fig. 7(b)	Fig. 7(c)	Fig. 7(d)	275	0.7
Si	Elastic	2329	2.6×10^{-6}	0.28	170	----	----
Cu_6Sn_5	Elastic	Fig. 8(a)	Fig. 8(b)	0.3	85.56	----	----
Cu_3Sn	Elastic	Fig. 8(c)	Fig. 8(d)	0.299	108	----	----
Co	Elastic	Fig. 9(a)	Fig. 9(b)	Fig. 9(c)	Fig. 9(d)	----	----
Poly-Si	Elastic	Fig. 10(a)	Fig. 10(b)	Fig. 10(c)	Fig. 10(d)	----	----
SiO_2	Elastic	Fig. 11(a)	Fig. 11(b)	Fig. 11(c)	Fig. 11(d)	----	----

and Cu_6Sn_5 were obtained from the literature [33]. The material parameters employed are shown in Table I. All other temperature-dependent material parameter plots for Co, SiO_2 , poly-Si, Cu_6Sn_5 , and Cu_3Sn are shown in the Appendix.

III. RESULTS AND DISCUSSION

A. Cross-Sectional Analysis

Fig. 3 shows diced bonded wafers and a chip's top view schematic, along with two cross-sectional SEM/EDX micrographs and a high-magnification bond image demonstrating the alignment accuracy of 500 nm. The TSV wafers were successfully bonded to the device (MEMS) wafers and diced with 100% yield. However, there is a small amount of squeeze out observed at the edge of the SLID bonds on the device wafer. This squeeze-out remains critical, especially for small microbumps with high density, as it needs to be controlled to prevent short-circuit failures. In this study, the metallization pad on the device wafers was intentionally designed to be

larger than the Cu–Sn–In SLID size. Nonetheless, in the squeezed-out area, the Co metallization was fully consumed, resulting in the formation of CoSn_3 and $(\text{Cu},\text{Co})_6(\text{Sn},\text{In})_5$ phases. Controlling squeeze out is still possible by adjusting various parameters such as bonding force.

Fig. 4 displays SEM/EDX mapping micrographs depicting the cap TSV poly-Si with SiO_2 isolation layer, along with the SLID bonding and Al metallization layer, crucial for facilitating the electrical path between TSVs. Phases across the bond line were identified by averaging measurements from at least five locations using EDX. The targeted single $(\text{Cu},\text{Co})_6(\text{Sn},\text{In})_5$ phase was successfully achieved, stacked between Cu and Co. Notably, no voids or cracks were observed in the bond line or within the Si, even at the edges of TSV-SLID, where are prone to stress formation and subsequently cracks. This stands in contrast to observations in the Cu–Sn SLID-TSV system used in previous works, as a part of the Eniac Project Lab4MEMS II. In that study, a bonding temperature of 320 °C was chosen to achieve a stable microstructure ($\text{Cu}-\text{Cu}_3\text{Sn}-\text{Cu}$)

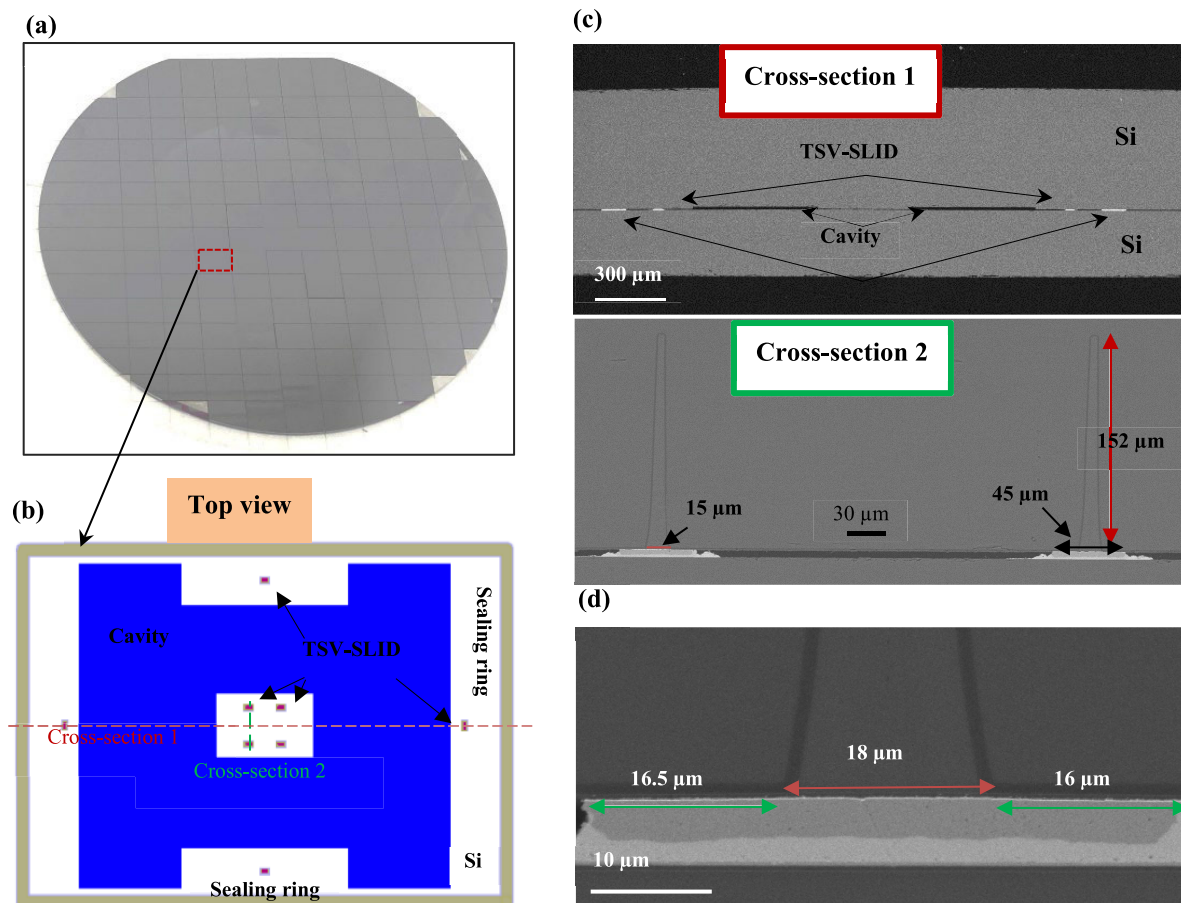


Fig. 3. (a) Diced bonded wafer, (b) chip's top view schematic, (c) two cross-sectional SEM/EDX micrographs, and (d) high-magnification bond images highlighting misalignments.

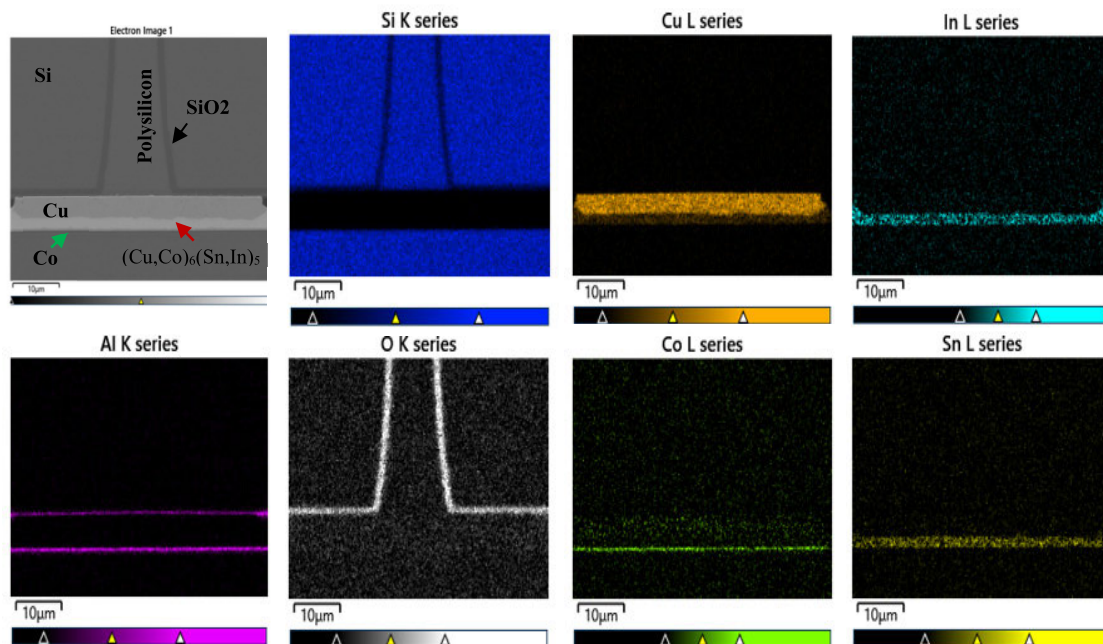


Fig. 4. SEM/EDX mapping micrographs illustrating the cap TSV poly-Si with SiO₂ isolated layer, as well as SLID bonding and Al metallization layer for the electrical path between TSVs.

[18]. In contrast, our current work utilized a reduced bonding temperature of 200 °C. This adjustment was based on the

incorporation of In into the system, which reduce the melting point. In addition, according to the literature, the stable

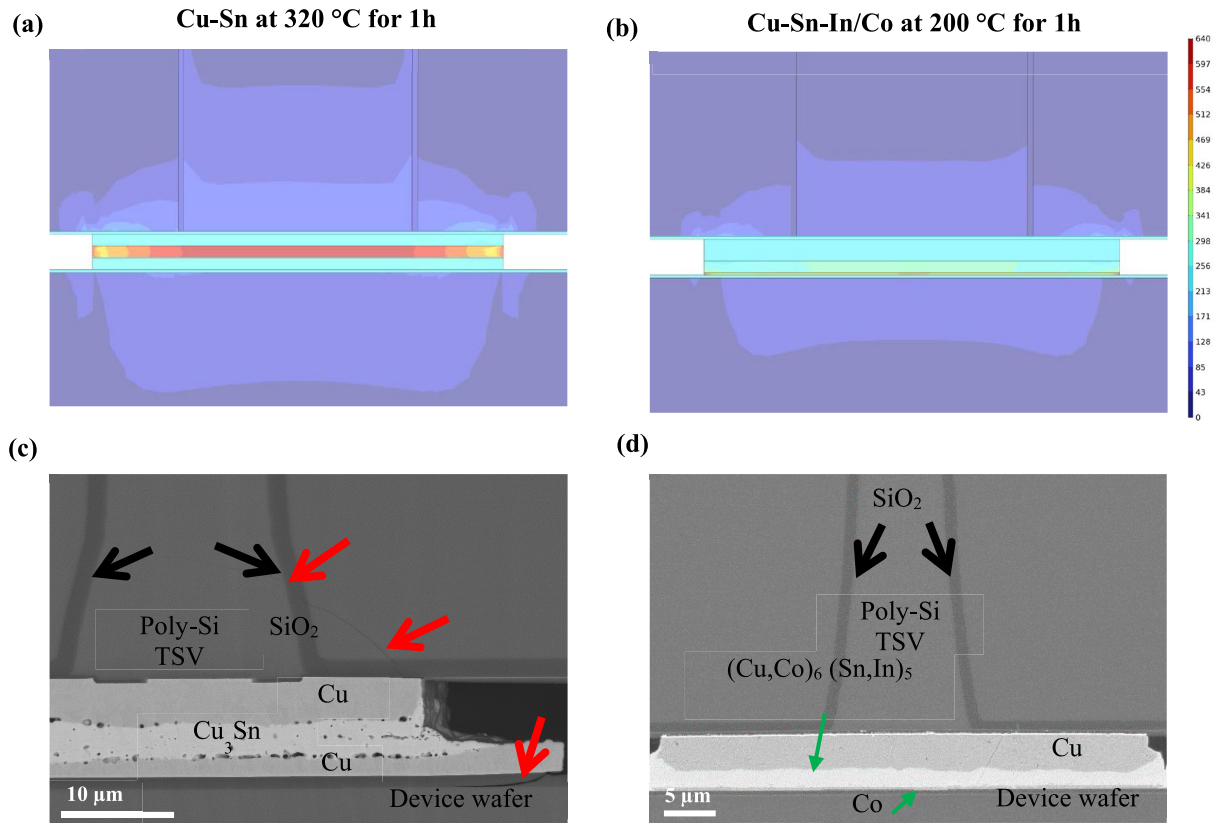


Fig. 5. von Mises stress plots for (a) Cu-Sn SLID-TSV bonded at 320 °C for 1 h, and (b) Cu-Sn-In/Co SLID-TSV bonded at 200 °C for 1 h, and the SEM/EDX cross-sectional micrograph of (c) Cu-Sn SLID-TSV obtained from the Eniac Project Lab4MEMS II, and (d) Cu-Sn-In/Co SLID-TSV obtained from the current study.

microstructure in Cu-Sn-In/Co is Cu-(Cu,Co)₆(Sn,In)₅-Co, eliminating the need to form Cu₃Sn as a stable phase. This offers the possibility of further lowering the bonding temperature. Research by N. Tiwary has demonstrated that in the SLID system, the ramp-down step significantly influences the overall stress level in the bonds. Thus, the primary contributor to the thermomechanical stresses in SLID bonds is the coefficient of thermal expansion (CTE) mismatch. An average decrease of over 30% in the von Mises stress level in Cu₃Sn bonds is observed when the bonding temperature is reduced from 320 °C to 200 °C [25]. Therefore, simulations were conducted to model the cooling down process of both the Cu-Sn and Cu-Sn-In/Co SLID-TSV systems while considering their stable microstructure. Fig. 5(a) and (b) illustrates the cross-sectional von Mises stress plots for both systems, along with the SEM/EDX cross section of the Cu-Sn SLID-TSV obtained from the Eniac Project Lab4MEMS II, and the Cu-Sn-In/Co SLID-TSV obtained from the current study. Similar phenomena, as observed by Tiwary et al. [25], were noted in the current work as well: higher bonding temperature resulted in higher thermomechanical stresses within the system. In the Cu-Sn system, the highest stresses were observed in the IMC area. However, in the Cu-Sn-In/Co SLID system, the highest stresses were found in the remaining Co layer. This issue could potentially be addressed by modifying the thickness of the Co layer, either by reducing its thickness or allowing it to be completely consumed in IMC formation. As can be seen

in Fig. 5(a) and (b), the von Mises stress is distributed at a greater depth in both Si and TSV wafers in the Cu-Sn system, with higher stress levels compared to the Cu-Sn-In/Co SLID system. Furthermore, the SEM/EDX micrograph of the Cu-Sn SLID-TSV [Fig. 5(c)] reveals cracks within the Si and TSV. Comparing the SEM/EDX image with the von Mises stress distribution suggests that this type of stress might cause crack formation and propagation along the stress distribution area. Si cracking due to high stress has also been observed in previous studies of Cu-Sn SLID-TSV bonding [18]. Conversely, no cracks were observed in the Cu-Sn-In/Co SLID-TSV system, which is a significant positive outcome of the new system. From the SEM micrograph of the Cu-Sn SLID-TSV [Fig. 5(d)], the presence of voids across the bond line within Cu₃Sn is clearly evident, a phenomenon that has been widely reported [34], [35]. Incorporating these voids into the simulation of Cu-Sn SLID-TSV could potentially result in even higher local stress levels [25]. Furthermore, these voids could serve as initiation points for cracks and subsequent propagation due to stress concentration across the voids, a well-known issue associated with thermal cycling that has been previously reported [17], [36]. In contrast, the new SLID-TSV system introduced here is void-free, eliminating concerns regarding the presence of potential crack initiation points and subsequent failure. Particularly, noteworthy is the stability of the Cu-Sn-In/Co SLID system, as reported by Emadi et al. [30]. According to their study, the system remains stable even

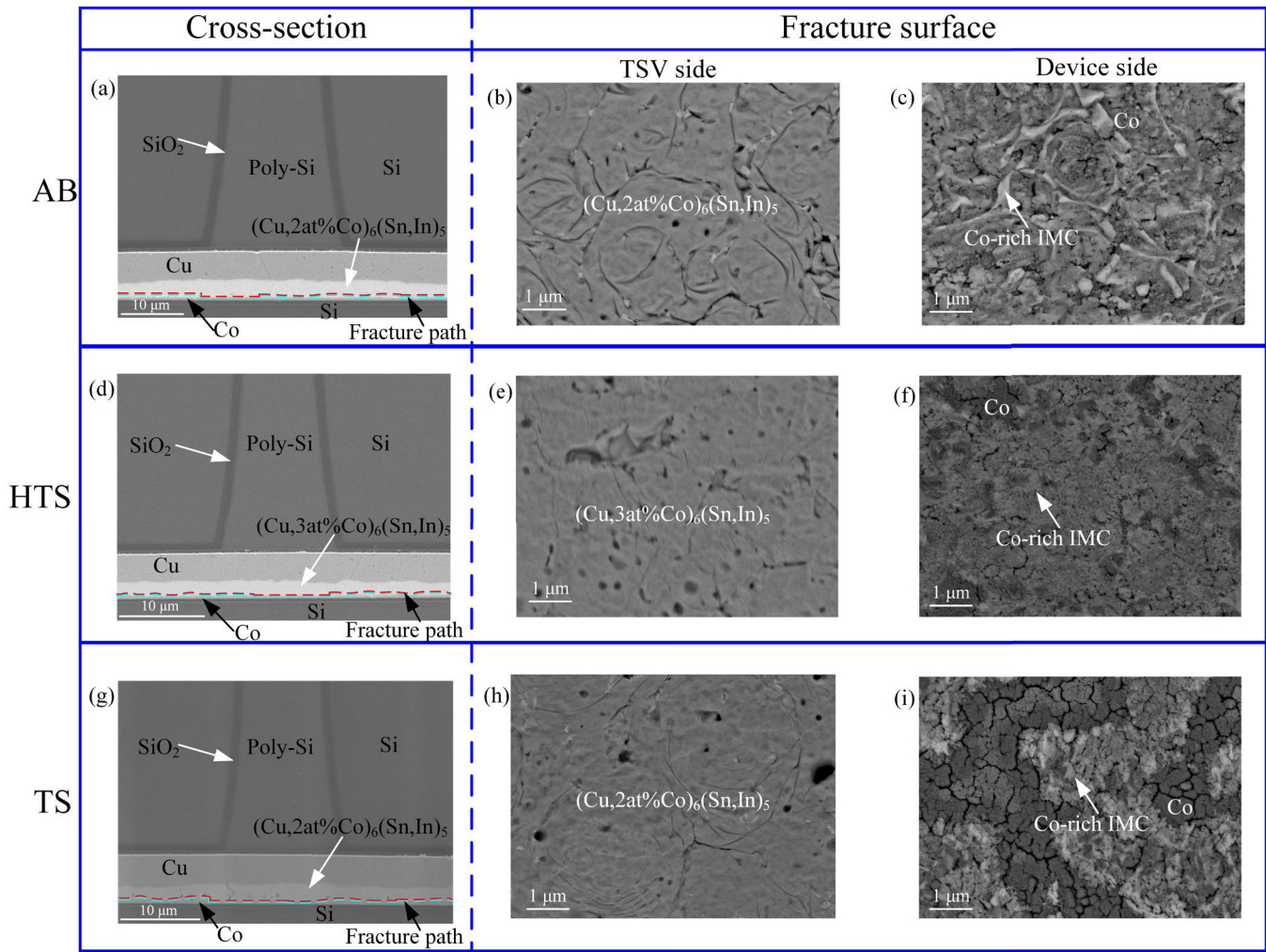


Fig. 6. SEM/EDX micrographs, cross-sectional, and fracture surfaces of AB Cu-Sn-In/Co SLID-TSV interconnects (a) cross-sectional image, (b) TSV side fracture surface, and (c) device side fracture surface. (d)–(f) HTS tested. (g)–(i) TS tested.

after prolonged annealing at 150 °C, with no formation of Cu_3Sn and subsequent voids.

B. TSV-SLID Thermomechanical Reliability

The Cu-Sn-In/Co SLID-TSV samples underwent HTS and TS tests. Following this, all samples (as-bonded, HTS treated, and TS tested) underwent tensile testing. Fig. 6 shows SEM/EDX cross-sectional micrographs and tensile fracture surfaces of the Cu-Sn-In/Co SLID-TSV interconnects after bonding (AB) and subsequent HTS and TS tests. Following the HTS and TS tests, no new phases were observed, and the bond line remained void- and crack-free, except for some of the TS-tested samples where a small number of voids and vertical cracks were observed. This could be attributed to the local CTE mismatch between metallization layers and IMC (as listed in Table II), as well as the TS applied to the joint during testing. Since there is no Cu present on one side of the joint to mitigate crack growth [37], and considering the relatively high CTE mismatch between Si, Co, and Cu_6Sn_5 , crack formation could potentially occur. However, only a small number of samples exhibited these cracks, with typically only a couple of cracks per sample. Previous research has

TABLE II
CTE VALUES

	Si	Cu	Co	Cu_6Sn_5
CTE (ppm/°C)	2.6[38]	16-18[38]	9-10[39]	16.3[40]

demonstrated that a higher number of voids in the bond line could increase the likelihood of crack formation, as observed in Cu-Sn SLID [17]. Nonetheless, in the Cu-Sn-In/Co SLID system, the void formation was minimized by preventing Cu_3Sn formation and subsequent void formation, resulting in a negligible number of cracks.

As can be seen in Fig. 6, the fracture paths were primarily located close to the Co layer. The fracture surfaces consisted of Co with a thin Co-rich phase in the device wafers and $(\text{Cu},\text{Co})_6(\text{Sn},\text{In})_5$ in the cap wafers. There is a slight variation in Co content from AB to HTS and TS samples, with the highest Co content observed in HTS samples, indicating a minor diffusion of Co into IMCs during HTS testing. These findings are consistent with previous research on Cu-Sn-In/Co joints (without TSVs) presented by Emadi et al. [30] and

TABLE III
TENSILE STRENGTH VALUES

Tested groups	Tensile strength (MPa)	Deviation
AB	10.67	1.60
TS	7.50	2.20
HTS	13.39	1.42

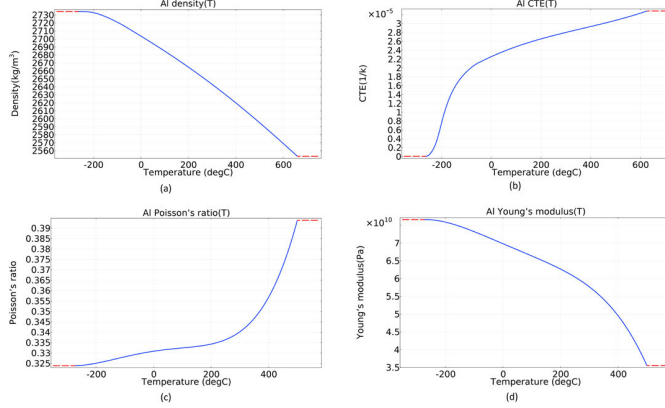


Fig. 7. Al temperature-dependent material data for (a) density, (b) CTE, (c) Poisson's ratio, and (d) Young's modulus.

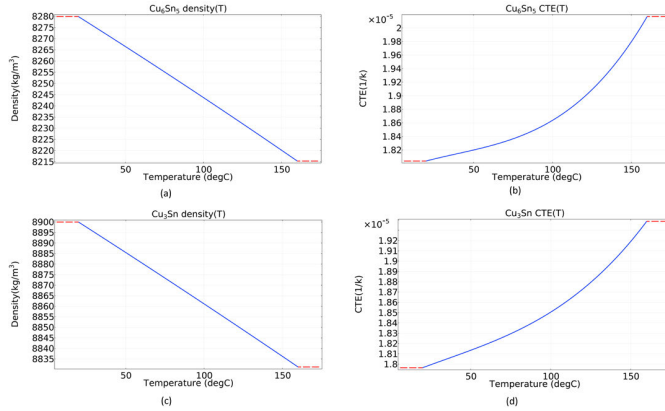


Fig. 8. Cu_6Sn_5 temperature-dependent material data for (a) density, (b) CTE and Cu_3Sn , (c) density, and (d) CTE.

indicate that integrating TSVs with SLID interconnects did not change the fracture surfaces or failure modes of Cu–Sn–In/Co SLID joints. Table III presents the tensile strength values for the LT-SLID-TSVs tested samples. All samples, AB, HTS-tested, and TS-tested samples exceeded the minimum requirements outlined in MIL-STD-883 method 2027.2. The highest value was observed after the HTS test. Potential reasons for this observation have been discussed in [30].

IV. CONCLUSION

This study successfully demonstrated the development of low-temperature 3-D interconnects using Cu–Sn–In/Co SLID, connected with TSVs, promising potential for 3-D MEMS-ICs integration. The interconnects showed a defect-free, crack-free microstructure, maintaining a stable $(\text{Cu},\text{Co})_6(\text{Sn},\text{In})_5$ single-phase composition even after undergoing HTS testing. However, a thin Co-rich phase was identified on the tensile fracture surfaces. Notably, no Cu_3Sn formation and subsequent void formation, common in Cu–Sn SLID systems, was

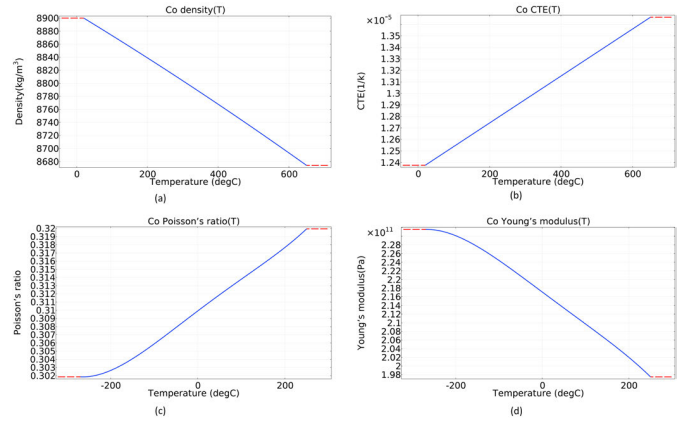


Fig. 9. Co temperature-dependent material data for (a) density, (b) CTE, (c) Poisson's ratio, and (d) Young's modulus.

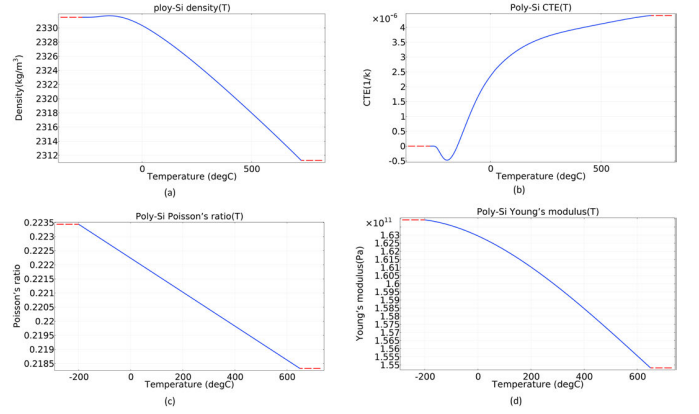


Fig. 10. Poly-Si temperature-dependent material data for (a) density, (b) CTE, (c) Poisson's ratio, and (d) Young's modulus.

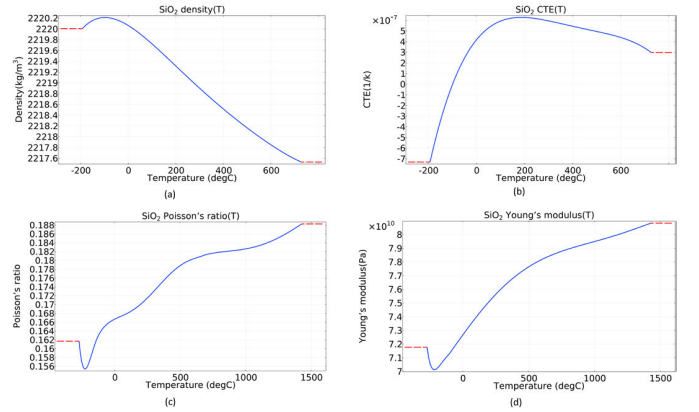


Fig. 11. SiO_2 temperature-dependent material data for (a) density, (b) CTE, (c) Poisson's ratio, and (d) Young's modulus.

observed, thereby potentially reducing the stress within the system, and minimizing the crack formation. Comparatively, considering the simulations, the proposed Cu–Sn–In/Co SLID system exhibited lower residual stress levels compared to Cu–Sn SLID-TSV configurations. In addition, the highest stresses in Cu–Sn–In/Co SLID-TSV and Cu–Sn SLID-TSVs are concentrated within the remaining Co layer and within the IMCs area, respectively. However, in simulations, void formation within Cu–Sn SLID joints was not considered, yet higher stress levels compared to the Cu–Sn–In/Co system

persisted due to the higher bonding temperature (320 °C). This phenomenon elucidates the crack formation observed in Cu–Sn TSV interconnects, emphasizing that adopting the Cu–Sn–In/Co system and lowering bonding temperature (200 °C) can significantly mitigate crack formations across bond lines, Si, and TSVs. In Cu–Sn–In/Co SLID-TSV configurations, TS testing revealed only minimal voids across bond lines in some samples, with a few cracks. Remarkably, all samples (including those with a few voids and cracks) met MIL-STD-883 method 2027.2 requirements for tensile strength testing. Notably, samples subjected to HTS testing exhibited the highest tensile strength values.

APPENDIX

See Figs. 7–11.

REFERENCES

- [1] Y. Grauer et al., “Overlay challenges in 3D heterogeneous integration,” *Proc. SPIE*, vol. 12053, p. 27, May 2022, doi: [10.1117/12.2608319](https://doi.org/10.1117/12.2608319).
- [2] S. Zhang et al., “Challenges and recent perspectives of 3D heterogeneous integration,” *e-Prime—Adv. Elect. Eng., Electron. Energy*, vol. 2, Jan. 2022, Art. no. 100052, doi: [10.1016/j.prime.2022.100052](https://doi.org/10.1016/j.prime.2022.100052).
- [3] Y.-G. Lee, M. McInerney, Y.-C. Joo, I.-S. Choi, and S. E. Kim, “Copper bonding technology in heterogeneous integration,” *Electron. Mater. Lett.*, vol. 20, no. 1, pp. 1–25, Jan. 2024.
- [4] N. P. Pham et al., “Zero-level packaging for (RF-)MEMS implementing TSVs and metal bonding,” in *Proc. IEEE 61st Electron. Compon. Technol. Conf. (ECTC)*, May 2011, pp. 1588–1595.
- [5] J. H. Lau, *Heterogeneous Integrations*. Singapore: Springer, 2019.
- [6] X. Li, J. Wang, M. Li, and W. Hou, “Low-temperature, short-time, wafer-level bonding for Cu/Sn/Cu solid-state-diffusion interconnect in 3-D integration,” *Phys. Scripta*, vol. 98, no. 2, Feb. 2023, Art. no. 025608, doi: [10.1088/1402-4896/acab8c](https://doi.org/10.1088/1402-4896/acab8c).
- [7] N. Lietaer, M. Taklo, K. Schjolberg-Henriksen, and P. Ramm, “(Invited) 3D interconnect technologies for advanced MEMS/NEMS applications,” *ECS Trans.*, vol. 25, no. 38, pp. 87–95, Apr. 2010, doi: [10.1149/1.3390661](https://doi.org/10.1149/1.3390661).
- [8] F. J. Blanco et al., “Novel three-dimensional embedded SU-8 microchannels fabricated using a low temperature full wafer adhesive bonding,” *J. Micromech. Microeng.*, vol. 14, no. 7, pp. 1047–1056, Jul. 2004, doi: [10.1088/0960-1317/14/7/027](https://doi.org/10.1088/0960-1317/14/7/027).
- [9] F. Niklaus, P. Enoksson, E. Kälvesten, and G. Stemme, “Low-temperature full wafer adhesive bonding,” *J. Micromech. Microeng.*, vol. 11, no. 2, pp. 100–107, Mar. 2001.
- [10] Y. Liu and K. N. Tu, “Low melting point solders based on Sn, Bi, and in elements,” *Mater. Today Adv.*, vol. 8, Dec. 2020, Art. no. 100115.
- [11] K.-K. Xu, L. Zhang, L.-L. Gao, N. Jiang, L. Zhang, and S.-J. Zhong, “Review of microstructure and properties of low temperature lead-free solder in electronic packaging,” *Sci. Technol. Adv. Mater.*, vol. 21, no. 1, pp. 689–711, Jan. 2020.
- [12] Y. Shi et al., “Fabrication of high-density micro-bump arrays for 3D integration of MEMS and CMOS,” *IEEE Trans. Electr. Electron. Eng.*, vol. 19, no. 5, pp. 785–790, Mar. 2024, doi: [10.1002/tee.24058](https://doi.org/10.1002/tee.24058).
- [13] E. Chery, C. Fohn, J. D. Messemacker, and E. Beyne, “Reliability challenges in advanced 3D technologies: The case of through silicon vias and SiCN–SiCN wafer-to-wafer hybrid-bonding technologies,” *IEEE Trans. Device Mater. Rel.*, vol. 23, no. 4, pp. 615–622, Dec. 2023.
- [14] J. D. Reed, M. Lueck, C. Gregory, A. Huffman, J. M. Lannon, and D. Temple, “High density interconnect at 10 μ m pitch with mechanically keyed Cu/Sn–Cu and Cu–Cu bonding for 3-D integration,” in *Proc. 60th Electron. Compon. Technol. Conf. (ECTC)*, Jun. 2010, pp. 846–852.
- [15] H. Xu et al., “Wafer-level SLID bonding for MEMS encapsulation,” *Adv. Manuf.*, vol. 1, no. 3, pp. 226–235, Sep. 2013, doi: [10.1007/s40436-013-0035-0](https://doi.org/10.1007/s40436-013-0035-0).
- [16] L. Sun, M.-H. Chen, and L. Zhang, “Microstructure evolution and grain orientation of IMC in Cu–Sn TLP bonding solder joints,” *J. Alloys Compounds*, vol. 786, pp. 677–687, May 2019.
- [17] A. Rautiainen, V. Vuorinen, J. Li, and M. Paulasto-Kröckel, “Vertical cracking of Cu–Sn solid-liquid interdiffusion bond under thermal shock test,” *Mater. Today, Proc.*, vol. 4, no. 7, pp. 7093–7100, 2017, doi: [10.1016/j.matpr.2017.08.002](https://doi.org/10.1016/j.matpr.2017.08.002).
- [18] V. Vuorinen, G. Ross, H. Viljanen, J. Decker, and M. Paulasto-Kröckel, “Process integration and reliability of wafer level SLID bonding for poly-Si TSV capped MEMS,” in *Proc. 7th Electron. System-Integr. Technol. Conf. (ESTC)*, Sep. 2018, pp. 1–6.
- [19] H. Liu, G. Salomonsen, K. Wang, K. E. Aasmundtveit, and N. Hoivik, “Wafer-level Cu/Sn to Cu/Sn SLID-bonded interconnects with increased strength,” *IEEE Trans. Compon., Packag., Manuf. Technol.*, vol. 1, no. 9, pp. 1350–1358, Sep. 2011, doi: [10.1109/TCPMT.2011.2156793](https://doi.org/10.1109/TCPMT.2011.2156793).
- [20] T.-T. Luu, A. Duan, K. E. Aasmundtveit, and N. Hoivik, “Optimized cu-sn wafer-level bonding using intermetallic phase characterization,” *J. Electron. Mater.*, vol. 42, no. 12, pp. 3582–3592, Dec. 2013, doi: [10.1007/s11664-013-2711-z](https://doi.org/10.1007/s11664-013-2711-z).
- [21] T.-T. Luu, A. Duan, K. Wang, K. E. Aasmundtveit, and N. Hoivik, “Optimization of Cu/Sn wafer-level bonding based upon intermetallic characterization,” in *Proc. 4th Electron. Syst.-Integr. Technol. Conf.*, Sep. 2012, pp. 1–5, doi: [10.1109/ESTC.2012.6542151](https://doi.org/10.1109/ESTC.2012.6542151).
- [22] J. Cai, J. Wang, Q. Wang, Z. Wu, and D. Wang, “Effects of current stress for low temperature Cu/Sn/Cu solid-state-diffusion bonding,” in *Proc. IEEE 67th Electron. Compon. Technol. Conf. (ECTC)*, May 2017, pp. 1742–1747, doi: [10.1109/ECTC.2017.96](https://doi.org/10.1109/ECTC.2017.96).
- [23] D. K. Mu, S. D. McDonald, J. Read, H. Huang, and K. Nogita, “Critical properties of Cu₆Sn₅ in electronic devices: Recent progress and a review,” *Current Opinion Solid State Mater. Sci.*, vol. 20, no. 2, pp. 55–76, Apr. 2016.
- [24] F. Somidin et al., “Characterising the polymorphic phase transformation at a localised point on a Cu₆Sn₅ grain,” *Mater. Characterization*, vol. 138, pp. 113–119, Apr. 2018, doi: [10.1016/j.matchar.2018.02.006](https://doi.org/10.1016/j.matchar.2018.02.006).
- [25] N. Tiwary, V. Vuorinen, G. Ross, and M. Paulasto-Kröckel, “Finite element simulation of Solid–Liquid interdiffusion bonding process: Understanding process-dependent thermomechanical stress,” *IEEE Trans. Compon., Packag., Manuf. Technol.*, vol. 12, no. 5, pp. 847–856, May 2022, doi: [10.1109/TCPMT.2022.3170082](https://doi.org/10.1109/TCPMT.2022.3170082).
- [26] F. Emadi, V. Vuorinen, G. Ross, and M. Paulasto-Kröckel, “Co, In, and Co–in alloyed Cu₆Sn₅ interconnects: Microstructural and mechanical characteristics,” *Mater. Sci. Eng., A*, vol. 881, Aug. 2023, Art. no. 145398, doi: [10.1016/j.msea.2023.145398](https://doi.org/10.1016/j.msea.2023.145398).
- [27] O. Golim et al., “Achieving low-temperature wafer level bonding with Cu–Sn–In ternary at 150 °C,” *Scripta Mater.*, vol. 222, Jan. 2023, Art. no. 114998.
- [28] F. Emadi, V. Vuorinen, S. Mertin, K. Widell, and M. Paulasto-Kröckel, “Microstructural and mechanical characterization of Cu/Sn SLID bonding utilizing Co as contact metallization layer,” *J. Alloys Compounds*, vol. 929, Dec. 2022, Art. no. 167228, doi: [10.1016/j.jallcom.2022.167228](https://doi.org/10.1016/j.jallcom.2022.167228).
- [29] F. Emadi, V. Vuorinen, and M. Paulasto-Kröckel, “Utilizing Co as a contact metallization for wafer-level Cu–Sn–In SLID bonding used in MEMS and MOEMS packaging,” in *Proc. IEEE 9th Electron. Syst.-Integr. Technol. Conf. (ESTC)*, Sep. 2022, pp. 359–363, doi: [10.1109/ESTC55720.2022.9939539](https://doi.org/10.1109/ESTC55720.2022.9939539).
- [30] F. Emadi, V. Vuorinen, S. Liu, and M. Paulasto-Kröckel, “Novel low-temperature interconnects for 2.5-/3-D MEMS integration: Demonstration and reliability,” *IEEE Trans. Compon., Packag., Manuf. Technol.*, vol. 14, no. 8, pp. 1337–1346, Aug. 2024, doi: [10.1109/tcpmt.2024.3430061](https://doi.org/10.1109/tcpmt.2024.3430061).
- [31] M. M. V. Taklo et al., “Residual stress in silicon caused by Cu–Sn wafer-level packaging,” in *Proc. Int. Tech. Conf. Exhib. Packag. Integr. Electron. Photonic Microsystems*, Jul. 2013, pp. 1–9, doi: [10.1115/1/PACK2013-73317](https://doi.org/10.1115/1/PACK2013-73317).
- [32] J. Tang, L. Li, G. Zhang, J. Zhang, and P. Liu, “Finite element modeling and analysis of ultrasonic bonding process of thick aluminum wires for power electronic packaging,” *Microelectron. Rel.*, vol. 139, Dec. 2022, Art. no. 114859, doi: [10.1016/j.microrel.2022.114859](https://doi.org/10.1016/j.microrel.2022.114859).
- [33] R. Fields. (2002). *Physical and Mechanical Properties of Intermetallic Compounds Commonly Found in Solder Joints*. [Online]. Available: http://www.metallurgy.nist.gov/mechanical_properties/solder_paper.html
- [34] K. E. Aasmundtveit, T.-T. Luu, K. Wang, and N. Hoivik, “Void formation in Cu–Sn solid-liquid interdiffusion (SLID) bonding,” in *Proc. Eur. Microelectron. Packag. Conf. (EMPC)*, Sep. 2015, pp. 1–6.
- [35] G. Ross, V. Vuorinen, and M. Paulasto-Kröckel, “Void formation in Cu–Sn micro-connects,” in *Proc. IEEE 65th Electron. Compon. Technol. Conf. (ECTC)*, May 2015, pp. 2193–2199.
- [36] A. Duan, K. Wang, K. E. Aasmundtveit, and N. Hoivik, “Reducing the bond frame width in Cu/Sn SLID wafer level packaging,” in *Proc. 4th Electron. System-Integr. Technol. Conf.*, Sep. 2012, pp. 1–5, doi: [10.1109/ESTC.2012.6542152](https://doi.org/10.1109/ESTC.2012.6542152).

- [37] H. Greve, S. Ali Moeini, P. McCluskey, and S. Joshi, "Prediction and mitigation of vertical cracking in high-temperature transient liquid phase sintered joints by thermomechanical simulation," *J. Electron. Packag.*, vol. 140, no. 2, pp. 1–9, Jun. 2018, doi: [10.1115/1.4039265](https://doi.org/10.1115/1.4039265).
- [38] M. Abtew and G. Selvaduray, "Lead-free solders in microelectronics," *Mater. Sci. Eng. R, Rep.*, vol. 27, nos. 5–6, pp. 95–141, Jun. 2000.
- [39] L. C. Briesse, J. Deubener, S. Selle, and T. Höche, "Thermal expansion of ϵ -Co and β -Co nanocrystals in glass matrices," *Mater. Lett.*, vol. 270, Jul. 2020, Art. no. 127707, doi: [10.1016/j.matlet.2020.127707](https://doi.org/10.1016/j.matlet.2020.127707).
- [40] D. R. Smith, T. A. Siewert, L. Stephen, and J. C. Madeni, "Database for solder properties with emphasis on new lead-free solders," Tech. Rep., 2022.



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